

ZeroQue® Purchase Control — Flow Diagram + Engineering Lock Addendum

1. Purchase Control Flow Diagram

```
flowchart TB
    subgraph CL[Client / Experience Layer]
        R[Requester / Store Ops]
        A[Approver / Finance]
        L[Logistics / Warehouse]
        N[Analyst / Search]
    end

    subgraph ADP[Application and Data Plane]
        subgraph P[Provisioning Service – System of Record]
            PR[Purchase Request]
            BA[Budget & Approval Check]
            PO[Purchase Order / PO Lines]
            OC[Order Charges]
            SA[Supplier Acknowledgement]
            FA[Fulfilment Acknowledgement]
            FC[Fulfilment Context\nnotes / images / location]
            PG[(Postgres\ncanonical truth)]
            OB[(Outbox Events)]
        end

        POL[Policy Service\nRBAC + policy\nallow / deny / require approval]
        G[Graph Service\ngovernance topology\nroles / line visibility\nLogisticsOperator scope]
        V[Vector Service\nindexed PO context\nmessages / notes / image metadata]
        I[Intelligence Service\nread-only summaries\ngoverned search\nanalytics]
    end

    subgraph EXT[External Systems / Fulfilment Context]
        S[Supplier / Email / API / EDI]
        E[ERP / AP / WMS Mirror]
        B[Blob Store\nimages]
        M[Messaging / Notifications]
        AI[AiFi / downstream integrations]
    end

    R --> PR
    PR --> BA
    BA --> POL
    POL -->|allow / require approval| A
```

```

A --> PO
PO --> OC
PO --> SA
SA --> S
S --> SA
SA --> FA
L --> FA
FA --> FC
FC --> B
FC --> M

PO --> PG
OC --> PG
SA --> PG
FA --> PG
FC --> PG
PG --> OB

OB --> G
OB --> V
G --> I
V --> I
N --> I

PO -. commercial mirror .-> E
SA -. event mirror .-> E
FA -. event mirror .-> E
AI -. downstream integration .-> FC

```

Boundary statement

ZeroQue owns commercial obligation, approval, PO issue, supplier acknowledgement, fulfilment acknowledgement, charges, and fulfilment context. ERP / WMS owns stock truth, warehouse movement, and accounting execution.

2. Engineering Lock Addendum — Purchase Control v1.0

2.1 Purpose

This addendum formalises how purchase requests, purchase orders, supplier acknowledgements, fulfilment acknowledgements, charges, and downstream ERP / WMS mirroring fit the existing Engineering Lock.

It does **not** change the core doctrine: - **Postgres** remains the sole system of record. - **Graph** remains a derived governance projection. - **Vector** remains a derived semantic projection. - **Intelligence** remains read-only and non-authoritative.

2.2 Core boundary

- **ZeroQue is the commercial control plane.**
- **ERP / WMS is the operational execution plane.**
- **ZeroQue records obligations and confirmations; ERP / WMS records stock and accounting.**
- **ZeroQue is canonical for purchase orders as commercial obligations; downstream systems may mirror those POs for operational and accounting purposes.**

2.3 What ZeroQue owns

ZeroQue owns the commercial transaction spine from intent through commitment and fulfilment confirmation: - purchase request - sourcing and supplier selection - budget checks - approval routing and approval decisions - purchase order issue - supplier acknowledgement - order charges such as carriage and small-order charges - fulfilment acknowledgement - fulfilment context such as notes, image evidence, timestamp, actor, and collection location

2.4 What ZeroQue does not own

ZeroQue does not become an ERP, inventory-management system, warehouse-management system, or accounts-payable engine.

It does **not** own: - stock ledgers - warehouse movements - bin locations - inventory valuation - goods receipt in the accounting or inventory sense - full three-way match - AP execution

2.5 Canonical object model

- **PurchaseRequest** — intent and justification before commitment.
- **PurchaseOrder** — canonical commercial obligation issued by ZeroQue after approval.
- **PurchaseOrderLine** — line-level commercial commitment with requester, department, cost-centre, and charge-allocation context.
- **SupplierAcknowledgement** — supplier-side accepted / partial / rejected / amended response.
- **FulfilmentAcknowledgement** — confirmation by the requester or an authorised logistics operator that the ordered line has been fulfilled to a degree sufficient to progress or close the commercial workflow.
- **OrderCharge** — carriage, small-order charge, or equivalent commercial charge attached to a PO and optionally allocated across lines.
- **FulfilmentContext** — light operational context such as handoff location, timestamp, actor, note, and image reference; useful for coordination but not inventory truth.

2.6 Normative rules

1. All purchase-control aggregates **MUST** live canonically in Postgres.
2. All projection writes to Graph and Vector **MUST** originate exclusively from Postgres outbox events.
3. ZeroQue **MUST** raise and issue the canonical commercial PO after approval.
4. ZeroQue **MUST NOT** maintain stock ledgers, bin-level inventory, inventory valuation, or warehouse movement state.
5. **FulfilmentAcknowledgement MUST NOT be treated as ERP goods receipt.**
6. ZeroQue **MAY** store fulfilment metadata such as location, image evidence, and contextual notes, provided that this metadata is not used to maintain inventory truth.
7. Downstream ERP / WMS systems **MAY** mirror purchase-control events and decide independently whether and how to create stock or accounting events.

2.7 Roles and visibility

- **Requester / Store Ops** — normally sees only their own lines and status; can submit requests and acknowledge fulfilment.
- **Approver / Finance** — sees governed approval context; can approve, reject, or escalate.
- **LogisticsOperator** — sees the full inbound PO for the authorised site / scope; can record arrival, add notes, attach image evidence, set collection location, and proxy-acknowledge on behalf of the requester.
- **Analyst / Search** — has read-only governed discovery through Graph, Vector, and Intelligence.

2.8 Recommended commercial state model

```
draft -> submitted -> approved -> issued -> supplier_acknowledged ->
partially_fulfilled / fulfilled -> closed
```

Side states may include: - rejected - cancelled - disputed

2.9 Minimum event contract

The minimum event set emitted through the existing outbox pattern should include: - `PurchaseRequestSubmitted` - `PurchaseRequestApproved` - `PurchaseOrderIssued` - `PurchaseOrderAmended` - `SupplierAcknowledged` - `OrderChargeAllocated` - `FulfilmentAcknowledged` - `PurchaseOrderClosed`

2.10 Mapping to Sourabh's current architecture

This proposal fits the current flow and architecture directly: - **Flow C** — Purchasing & Approval Workflows is already the correct home for `PurchaseRequest`, `PurchaseOrder`, `SupplierAcknowledgement`, `FulfilmentAcknowledgement`, `OrderCharge`, and `FulfilmentContext`. - **Flow D** — extend the current runtime flow from `approved -> PO issued -> spending event` to `approved -> PurchaseOrderCreated -> PurchaseOrderIssued -> SupplierAcknowledged -> FulfilmentAcknowledged -> Closed`. - **Flow E** — `PurchaseOrderIssued`, `SupplierAcknowledged`, and `FulfilmentAcknowledged` become first-class outbox events consumed by Graph, Vector, and Intelligence. - **Graph** remains governance-only: line ownership, role scope, approved universe, and LogisticsOperator visibility are graph-native concepts, but the PO itself remains canonical in Postgres. - **Vector** remains retrieval-only: PO notes, message context, and image metadata may be indexed for governed search, but never become authority.

2.11 Air Traffic Control anchor

This addendum remains fully consistent with the control-tower metaphor: - a **Purchase Order** is the commercial clearance to move - a **FulfilmentAcknowledgement** records that the authorised movement completed commercially - ERP / WMS still handles runway, cargo, and ground-operation truth

2.12 Draft insertion block for the Engineering Lock

Use the following language as an insertion block:

ZeroQue is the commercial control plane for purchase control. It is canonical for purchase orders as commercial obligations. ERP / WMS remains canonical for stock truth, warehouse movement, and accounting execution. ZeroQue records obligations and confirmations; ERP / WMS records stock and accounting. FulfilmentAcknowledgement in ZeroQue MUST NOT be interpreted as inventory goods receipt. Purchase-control aggregates MUST be stored in Postgres and projected through the existing outbox pattern into Graph and Vector.